Solving multi-step equations

1 Consider the equation $9\left(n+5\right)=4n+50$ for $n=1$.

a) Find the value of the left-hand side of the equation when $n=1$.

b) Find the value of the right-hand side of the equation when $n=1$.

c) Consider your answers from parts (a) and (b).
Is $n=1$ a solution of $9\left(n+5\right)=4n+50$?

2 Consider the equation $0.28n-0.4=2.3n-3.03$ for $n=0.8$.

a) Find the value of the left-hand side of the equation when $n=0.8$.

b) Find the value of the right-hand side of the equation when $n=0.8$.

c) Consider your answers from parts (a) and (b).
Is $n=0.8$ a solution of the equation $0.28n-0.4=2.3n-3.03$?

3 Determine whether the given value of the variable is a solution to the equation.

a) $p=0;\, 3p-9=7p-1$

b) $x=-2;\, 3x-\left(4x-8\right)=-6$

c) $x=\dfrac{7}{12};\, x-\dfrac{4}{9}=\dfrac{17}{36}$

d) $x=-3;\, \dfrac{-24}{x}+7=5x$

e) $x=\dfrac{1}{4};\, x+\dfrac{2}{3}=\dfrac{11}{12}$

4 Complete the following equation such that $x=-7$ is the only solution.
$2\left(x-12\right)+9=3x-\square$

5 Complete the following equation such that it has no solutions.
$80x+\dfrac{1}{6}=\square\left(8x+2\right)$

6 Solve the following equations:

a) $8k+2=15k-12$

b) $8r-2=4r+14$

c) $10r+8=74-r$

d) $12r+8=116-6r$

e) $36-2k=6k+20$

f) $-90-8v=-17v+9$

g) $\dfrac{14k-76}{2}-2=2k$

h) $-n-\dfrac{1}{3}=2$

i) $\dfrac{y-4.3}{-4}=7.7$

7 Solve for x:

a $6x+9=10x-3$

b $-\frac{6x}{5}+5=35$

c $\frac{11x+22}{8}=33$

d $3x-8=x$

e $5x+2=3x+22$

f $-107-8x=-17x+19$

g $\frac{5x}{6}+4=21.5$

h $6x-\frac{11}{7}=\frac{409}{7}$

i $\frac{11.8x+31.86}{4}=-106.2$

j $\frac{4x-20}{7}=\frac{28}{3}$

k $x-\frac{1}{5}=\frac{3}{4}$

l $7x-\frac{90}{7}=x$

m $8x-\frac{112}{3}=4x$

n $x-\frac{5}{12}=-\frac{1}{4}$

o $12x+19=x+85$

p $9x+4=10x+2$

8 Find the value of x.

a $3(x+1)+2=-1$

b $-2(x+2)-4=4$

c $3(x+6)+3(x+24)=12$

d $2(2x+5)=3(x+5)$

e $-7(x+1)=-28$

f $3(x-1)-5=-8.6$

g $3.8(x+4)-2=1.8$

h $3\left(x+\frac{3}{2}\right)+3=-\frac{15}{2}$

i $5.3(3x+55)=4.9(x+55)$

j $4(5x+1)=-3(5x-5)$

k $5(2x+2)=-3(4x-5)+5x$

9 Determine the number of solution each of the following equations has.

a $18+3x=3x-16$

b $8x-4x=4$

c $x+8=x+6$

d $2x-2=2x-2$

e $7x+4=6x+5$

f $8x=8x+8$

g $6x+5=7x-5$

h $9x-9=9x+7$

i $7x+7=9x-7$

j $2x=2x+4$

l $10(5+x)=10(x+5)$

m $\dfrac{10+x}{10}=\dfrac{x+4}{10}$

n $7(7+x)=2x+28$

o $-31-3(x-9)=2(x-2)-5x$

p $24+7(x-23)+3=16+8(x+14)-x-262$

10 Consider the equation $3x-29=9x+13$.

a Solve for x.

b How many solutions does the equation $3x-29=9x+13$ have?

11 For each of the following relations:

i Using n to represent the unknown number, form an equation that represents this relation.

ii Determine the number of solutions the relationship has.

a Three more than three times a number is equal to six more than four times the number

b Six more than two times a number is equal to five more than two times the number

c Two more than four times a number is equal to two added to quadruple the number